

Lesson 3 · Rebuild RAG with LangChain

PDF: [this lesson](#) · **Install (Linux · macOS · Windows):** [guide](#) · [PDF](#)

Part of [local-ai-lab](#) — a hands-on course for building local AI.

Course home: <https://nikolareljin.github.io/local-ai-lab/> **Source:** <https://github.com/nikolareljin/local-ai-lab>

Lessons: [1 · RAG](#) → [2 · MCP](#) → **3 · LangChain (you are here)** → [4 · LangGraph](#) → [5 · Ollama tools](#) → [6 · Semantic Kernel](#) → [7 · Bedrock Agents](#) → [8 · Google ADK](#)

Status: planned. Outline below; full step-by-step coming later. ☐ the repo to follow along.

The idea

In [Lesson 1](#) you built every RAG primitive by hand: a loader, a splitter, a retriever, a prompt, and a provider abstraction. **LangChain** ships all of those as components. This lesson rebuilds the **same** document Q&A app with LangChain — and then asks the honest question: **what did the framework buy us, and what did it cost?**

The goal is not to crown a winner. It's that, having written the primitives yourself, you can now read LangChain and see **exactly** which of your hand-rolled pieces each component replaces.

What you'll learn

- **Document loaders & text splitters** — LangChain's `PyPDFLoader`, `RecursiveCharacterTextSplitter` vs. your `extract.py` / `chunk.py`
- **Vector stores & retrievers** — Chroma / FAISS + `as_retriever()` vs. your BM25/embeddings
- **Chat models & embeddings** — `ChatOllama`, `ChatOpenAI`, `init_chat_model` vs. your provider adapters (the abstraction you already understand)
- **LCEL** — composing a retrieval chain with the LangChain Expression Language (`|` pipes)
- **Prompt templates** — `ChatPromptTemplate` vs. your `prompts.py`
- **Trade-offs** — abstraction and ecosystem vs. transparency, dependencies, and version churn

Maps directly onto Lesson 1

You built by hand (Lesson 1)	LangChain component (Lesson 3)
<code>extract.py</code>	DocumentLoaders (<code>PyPDFLoader</code> , <code>Docx2txtLoader</code> , ...)
<code>chunk.py</code>	<code>RecursiveCharacterTextSplitter</code>

<code>store.py</code> index	<code>VectorStore</code> (Chroma, FAISS)
<code>Bm25Retriever</code> / <code>EmbeddingRetriever</code>	<code>BM25Retriever</code> , <code>vectorstore.as_retriever()</code>
<code>prompts.py</code>	<code>ChatPromptTemplate</code>
<code>providers/</code>	<code>ChatModel</code> adapters (<code>ChatOllama</code> , <code>ChatOpenAI</code> , ...)
<code>engine.answer_question()</code>	an LCEL retrieval chain

The payoff of doing it the hard way first: none of the table above is mysterious. You know what every component does because you wrote its equivalent.

Prerequisites

Finish [Lesson 1](#). Bring the same `documents/` corpus — we'll get the same answers, built a different way.

Next lesson

Lesson 4 · LangGraph → — turn the chain into a stateful agent graph.

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